

MediQuery: A Domain-specific Multi-Agent Retrieval-Augmented Generation System for Biomedical Research Assistance

Jaya Prakash Yadav Gorla

Jarvis College of Computing and Digital Media
DePaul University
Chicago, United States
jgorla@depaul.edu

Kunal Jatin Tamhane

Jarvis College of Computing and Digital Media
DePaul University
Chicago, United States
ktamhane@depaul.edu

Abstract—The exponential growth of biomedical literature imposes a substantial cognitive burden on clinicians and researchers, who must synthesize evidence from large volumes of publications to support evidence-based decision-making [1]. General-purpose large language models (LLMs) partially address this need but remain prone to hallucination, lack real-time grounding, and provide limited citation transparency [3], [4]. We present MediQuery, a domain-specific multi-agent Retrieval-Augmented Generation (RAG) system designed for biomedical research assistance. MediQuery orchestrates three specialized retrieval agents: a local vector-database RAG agent powered by PubMedBERT embeddings in QdrantDB [7], [13], a live PubMed agent querying the NCBI E-utilities API, and a web search agent providing supplementary context under a LangGraph Supervisor that synthesizes heterogeneous evidence before invoking the generative model [12]. To further improve biomedical grounding, we fine-tuned LLaMA-3 8B using parameter-efficient adaptation techniques [9] on the PubMedQA corpus [10] and deployed the optimized model for local inference. We evaluated MediQuery through an ablation study across multiple retrieval configurations on the PubMedQA evaluation set. Results demonstrate that combining heterogeneous evidence sources improves semantic grounding and produces more reliable biomedical responses. An interactive Streamlit interface presents grounded, citation-traced responses alongside a dynamically constructed knowledge graph.

Index Terms—biomedical question answering, retrieval-augmented generation, multi-agent systems, LangGraph, PubMedBERT, LLaMA-3, LoRA, QdrantDB, knowledge graph, AWS, Docker, evidence-based medicine

I. INTRODUCTION

Biomedical literature continues to grow at a pace that makes it increasingly difficult for clinicians and researchers to remain current with emerging evidence [1]. The process of identifying relevant studies, synthesizing findings, and translating evidence into actionable insight introduces substantial cognitive overhead and can delay evidence-based decision-making.

Recent advances in large language models (LLMs) have created new opportunities for biomedical question answering and literature assistance. However, general-purpose models remain limited by hallucinations, outdated knowledge, and insufficient citation transparency, making them unreliable for high-stakes

biomedical applications [3], [4]. Retrieval-Augmented Generation (RAG) has emerged as a promising solution by grounding responses in external knowledge sources [3], but conventional single-source retrieval approaches often struggle with complex biomedical queries that require integrating information from multiple evidence channels [4].

To address these limitations, we propose *MediQuery*, a domain-specific multi-agent RAG framework for biomedical research assistance. MediQuery combines local semantic retrieval using PubMedBERT embeddings and QdrantDB [7], [13], live retrieval from PubMed, and supplementary web-based evidence under a LangGraph Supervisor that coordinates retrieval and response generation [12]. To improve domain adaptation, the generative component is further refined using biomedical question-answering data and deployed for efficient local inference [9].

We evaluate MediQuery through an ablation study across multiple retrieval configurations and demonstrate that integrating heterogeneous retrieval sources improves response grounding and overall answer quality. In addition to retrieval and generation, MediQuery provides an interactive interface for exploring generated responses with citation support and knowledge graph visualization.

The main contributions of this work are as follows:

- A multi-agent RAG architecture that integrates local biomedical retrieval, live PubMed search, and web-based evidence aggregation.
- A domain-adapted generation pipeline optimized for biomedical question answering.
- An evaluation framework demonstrating the benefits of heterogeneous retrieval strategies.
- An interactive system for citation-grounded response exploration and biomedical knowledge visualization.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system architecture. Section IV describes methodology and implementation details. Section V outlines the experimental setup. Section VI presents results. Section VII discusses findings. Section VIII outlines future work. Section IX concludes the paper.

II. RELATED WORK

A. Retrieval-Augmented Generation

Retrieval-Augmented Generation has emerged as an effective approach for improving factual grounding by augmenting language models with externally retrieved knowledge. Prior work demonstrated that retrieval-based generation improves performance on knowledge-intensive tasks and reduces hallucination compared to standalone generation models [3]. Subsequent research explored hybrid retrieval pipelines and reranking strategies to improve retrieval quality and robustness [6]. However, recent studies have identified several failure modes in production RAG systems, including retrieval insufficiency, evidence inconsistency, and unreliable generation under low-confidence retrieval conditions, motivating more adaptive retrieval architectures [4].

B. Biomedical NLP and Domain Adaptation

Domain-specific language models have become foundational to biomedical natural language processing. PubMedBERT demonstrated that pretraining exclusively on biomedical corpora improves downstream biomedical understanding relative to general-domain encoders [7]. Biomedical NLP frameworks such as SciSpaCy further enable scalable extraction of domain entities and relationships for downstream reasoning and knowledge representation tasks [8]. Parameter-efficient adaptation methods such as LoRA have also made it feasible to specialize large language models for domain-specific tasks while reducing computational requirements [9].

C. Biomedical Question Answering Benchmarks

Biomedical question answering benchmarks provide standardized evaluation for domain-adapted language systems. PubMedQA introduced expert-annotated biomedical question answering grounded in scientific literature and remains widely adopted for training and evaluation of biomedical QA models [10]. Additional datasets expanded evaluation beyond answer correctness to include evidence alignment and explanation quality, reflecting the growing emphasis on trustworthy biomedical generation [11].

D. Multi-Agent and Agentic LLM Systems

Recent advances in agentic LLM systems enable decomposition of complex tasks into coordinated workflows across specialized agents. Frameworks such as LangGraph support structured orchestration through explicit execution graphs and shared state management [12]. Multi-agent approaches have shown improvements in reasoning quality and task specialization in medical and scientific domains [15]. However, existing systems often rely on homogeneous retrieval pipelines and provide limited mechanisms for retrieval validation and evidence control. MediQuery extends this direction through heterogeneous retrieval orchestration and confidence-aware generation designed for biomedical evidence synthesis.

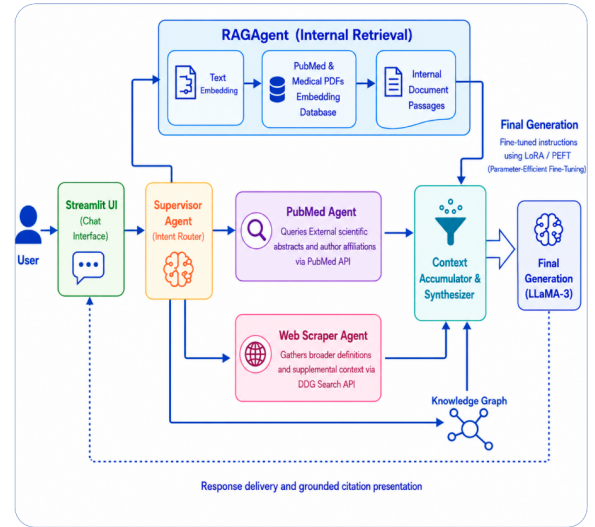


Fig. 1. MediQuery system architecture illustrating the six-layer pipeline consisting of the Streamlit interface, LangGraph Supervisor, retrieval agents, knowledge graph layer, LLM inference layer, and persistent storage components.

III. SYSTEM ARCHITECTURE

Figure 1 presents the MediQuery architecture, illustrating the end-to-end pipeline from user interaction through multi-agent retrieval, evidence synthesis, knowledge augmentation, generation, and storage. The system is designed around a strict separation of retrieval, reasoning, and response generation to improve grounding and citation traceability.

The user interaction layer is implemented through a Streamlit interface [11] that accepts natural language queries and optional PDF uploads. Uploaded documents are processed through a document ingestion pipeline, segmented into searchable chunks, embedded using PubMedBERT [7], and indexed into QdrantDB [13] to support retrieval in future queries. Additional interface components support document exploration and comparative analysis. The orchestration layer is managed by a LangGraph Supervisor [12] responsible for coordinating execution across retrieval agents, maintaining shared workflow state, and enforcing response guardrails. The Supervisor aggregates retrieved evidence, constructs the synthesis prompt, and invokes generation only when sufficient supporting evidence is available. When retrieval confidence is insufficient, the system returns an evidence-aware fallback response rather than generating unsupported output. The retrieval layer consists of three specialized agents. The local RAG agent retrieves semantically relevant passages from QdrantDB [13] using PubMedBERT embeddings [7] and diversity-aware reranking. The biomedical literature agent retrieves evidence from external scientific sources including PubMed and related repositories. The web search agent supplements indexed evidence with broader contextual information while preserving source transparency during synthesis. The knowledge augmentation layer constructs a biomedical knowledge graph using SciSpaCy [8] and NetworkX by extracting entities and relationships from

retrieved evidence. The graph provides structured contextual signals during generation and is visualized interactively within the user interface. The generation layer combines a domain-adapted LLaMA-3 model for response synthesis with retrieval representations derived from PubMedBERT [7]. Evidence collected across retrieval agents is consolidated into a grounded generation context before producing the final response. The storage layer maintains vector embeddings, document artifacts, retrieval corpora, and fine-tuning datasets required to support indexing, retrieval, and model adaptation throughout the system lifecycle.

IV. METHODOLOGY

A. Dataset

MediQuery integrates multiple biomedical datasets that serve complementary roles across retrieval, domain adaptation, and evaluation. To support evidence-grounded response generation, separate datasets were selected for corpus construction, supervised fine-tuning, and final system evaluation. Table I summarizes the datasets used throughout the pipeline.

TABLE I
DATASETS USED FOR RETRIEVAL, FINE-TUNING, AND EVALUATION IN
MEDIQUERY

Category	Source / Dataset	Size / Count
Corpus Data	PubMed	250 papers
	arXiv	125 papers
	medRxiv	125 papers
	Total Corpus Size	500 papers
Fine-Tuning Data	MedQUAD	14,984 QA pairs
	MedAESQA	40 QA pairs
Evaluation Data	PubMedQA	1,000 records

The retrieval corpus was assembled from biomedical literature sources to provide broad domain coverage and support evidence retrieval during inference. MedQUAD [18] served as the primary supervised fine-tuning dataset due to its broad coverage of medical question-answer scenarios, while MedAESQA [11] supplemented training with evidence-aware examples to encourage grounded response generation. PubMedQA [10] was reserved exclusively for evaluation and excluded from training and validation stages to ensure unbiased assessment of system performance. The live biomedical corpus was assembled to support retrieval grounding and expose the system to contemporary biomedical terminology and literature structure.

B. Preprocessing and Vector Indexing

Biomedical documents and user-uploaded PDFs were processed through a standardized ingestion and indexing pipeline designed to support scalable retrieval-based generation. To improve durability and decouple document ingestion from

downstream embedding and indexing workloads, the pipeline was deployed using Docker containers with QdrantDB [13] as the persistent vector storage layer and a Kafka-backed asynchronous queue for ingestion orchestration. Documents collected from heterogeneous biomedical sources were normalized into a unified chunk representation before indexing. The ingestion pipeline handled variations in source structure and formatting across biomedical literature repositories and incorporated fault-tolerant deserialization mechanisms to improve robustness against incomplete metadata, malformed records, and noisy PDF extraction artifacts commonly encountered in real-world documents. Raw documents were segmented into manageable text chunks to preserve semantic coherence while maintaining retrieval efficiency. Across the collected biomedical corpus, the ingestion workflow processed approximately 500 papers and produced more than 8,500 vector-indexed chunks for retrieval. Each text segment was transformed into a dense vector representation using PubMedBERT [7], a domain-specific biomedical encoder trained on PubMed literature. Dense embeddings were selected to enable semantic retrieval beyond exact keyword matching and to improve relevance for complex biomedical queries. Generated embeddings were indexed within QdrantDB [13] to support low-latency similarity search during inference. Retrieved passages were subsequently filtered and reranked to reduce redundancy and improve evidence diversity before downstream synthesis. To support personalized retrieval, a separate indexing workflow was maintained for user-uploaded documents. Uploaded PDFs were dynamically processed and inserted into a dedicated retrieval collection, enabling immediate incorporation into subsequent user queries while preserving separation from the core biomedical corpus.

C. Fine-Tuning Pipeline

LLaMA-3 8B Instruct was selected as the generative backbone for biomedical response synthesis due to its strong instruction-following capabilities and suitability for local deployment. To adapt the model for biomedical question answering while maintaining computational efficiency, parameter-efficient fine-tuning was performed using Low-Rank Adaptation (LoRA) [9]. Fine-tuning was conducted using MedQUAD [18] as the primary supervised training dataset and supplemented with MedAESQA [11] to introduce evidence-aware biomedical reasoning examples. Training inputs were structured using an instruction-based conversational format and enriched with biomedical context to encourage grounded response generation. During adaptation, only lightweight trainable adapter parameters were updated while preserving the original model weights. This approach reduced computational requirements and minimized catastrophic forgetting while enabling specialization toward biomedical language and question-answering behavior [9]. Following training, the adapted model was exported in an optimized format for local inference and integrated into the MediQuery generation pipeline. The resulting deployment supports efficient biomed-

ical response synthesis without requiring continuous cloud-based model hosting [14].

D. Multi-Agent Orchestration

MediQuery adopts a multi-agent orchestration strategy to coordinate retrieval and generation across heterogeneous biomedical evidence sources. The orchestration layer is implemented using LangGraph [12] and operates as a structured workflow that separates retrieval, evidence aggregation, synthesis, and response generation into independent stages. The retrieval stage consists of three specialized agents with complementary responsibilities. The local RAG agent performs semantic retrieval from the indexed biomedical corpus and user-uploaded documents using dense vector search. The biomedical literature agent retrieves recent scientific evidence from external repositories to supplement locally indexed knowledge. The web search agent provides broader contextual information and definitions to support queries that extend beyond curated biomedical datasets. Retrieved evidence from all agents is aggregated by a supervisory synthesis stage that consolidates heterogeneous information into a unified generation context. Evidence sources are explicitly separated during synthesis to preserve provenance and maintain transparency across retrieval channels. To reduce unsupported generation, MediQuery incorporates a retrieval-confidence circuit breaker prior to inference. Generation proceeds only when sufficient supporting evidence is available across one or more retrieval sources. When evidence quality is insufficient, the system returns an evidence-aware fallback response rather than generating speculative biomedical content. This orchestration framework enables MediQuery to combine local retrieval efficiency, access to recent biomedical literature, and broader contextual grounding while maintaining traceability and response reliability.

E. Knowledge Graph Module

To enhance contextual understanding and improve interpretability, MediQuery constructs a dynamic biomedical knowledge graph from retrieved evidence. The knowledge graph serves as an intermediate structured representation that complements retrieval-based generation by organizing biomedical concepts and their relationships. Medical entities and relationships are extracted from retrieved passages using biomedical natural language processing techniques supported by SciSpaCy [8]. Extracted entities include concepts such as diseases, drugs, methods, and related biomedical terminology, which are represented as interconnected nodes and edges within the graph structure. The generated knowledge graph is incorporated into the response generation pipeline as structured contextual information. By exposing relationships between entities rather than relying solely on retrieved passages, the graph supports richer contextual grounding and enables exploration of related biomedical concepts during synthesis. To improve usability and transparency, the knowledge graph is rendered interactively within the user interface, allowing users to inspect extracted entities and navigate relationships associated with the generated response. The graph is updated

dynamically throughout user interaction to reflect newly retrieved evidence.

F. User Interface

MediQuery provides an interactive interface for biomedical question answering and evidence exploration. Users can submit natural language queries and optionally upload documents to support personalized retrieval. The interface integrates literature retrieval, grounded response generation, and knowledge graph visualization within a unified workflow. Retrieved evidence is organized by provenance to improve transparency and allow users to inspect supporting sources associated with generated responses. Additional interface capabilities support document comparison and corpus-level summarization to facilitate exploration of biomedical literature.

V. EXPERIMENTS AND EVALUATION

A. Experimental Setup

MediQuery was evaluated using the PubMedQA benchmark [10] to assess its ability to generate grounded biomedical responses under different retrieval conditions. Experiments were conducted on a held-out subset of expert-labeled evaluation samples selected from the benchmark dataset.

To isolate the effect of retrieval on response quality, all configurations used the same generative model and identical evaluation inputs. Only retrieval availability was varied across experiments.

As this work focuses on demonstrating the feasibility of a multi-agent biomedical RAG framework, experiments were conducted at prototype scale. Larger-scale evaluation across the complete benchmark dataset is reserved for future work.

B. Evaluation Metrics

System performance was measured using automatic lexical and semantic similarity metrics commonly used in text generation and biomedical question answering.

- **BLEU-1:** Measures unigram overlap and evaluates lexical precision [16].
- **ROUGE-1:** Measures recall-oriented overlap and evaluates content coverage [17].
- **METEOR:** Incorporates stemming and synonym matching to provide a more flexible lexical assessment [18].
- **BERTScore F1:** Measures semantic similarity using contextual embeddings and captures meaning preservation beyond exact word overlap [19].

Metric computation was performed consistently across all experimental configurations.

C. Ablation Study Design

To evaluate the contribution of retrieval, MediQuery was tested under three experimental configurations.

- **Baseline:** Retrieval disabled; responses generated directly from the query.
- **RAG-Only:** Retrieval limited to the local vector database.

- **Full Multi-Agent:** Retrieval enabled across local retrieval, biomedical literature retrieval, and supplementary contextual sources.

Each configuration was evaluated on the same benchmark samples to ensure controlled comparison.

D. Baseline Comparison

The baseline configuration served as the primary reference for evaluating retrieval effectiveness. By keeping the generative model fixed and varying only retrieval availability, improvements in performance can be attributed to evidence grounding rather than changes in generation capability.

E. Ablation Study Results

This study summarizes performance across the three retrieval configurations.

The results demonstrate that retrieval substantially improves response quality compared to generation without external evidence. The Baseline configuration produced lower lexical and semantic alignment scores, indicating that responses generated without retrieval grounding are less consistent with reference answers.

The RAG-Only configuration improved lexical precision, achieving the highest BLEU-1 score among all settings. This suggests that local semantic retrieval supports generation of responses with stronger surface-level alignment to reference text.

The Full Multi-Agent configuration achieved the strongest overall performance, obtaining the highest ROUGE-1 and BERTScore F1 scores. The improvement in ROUGE-1 indicates better content coverage, while the increase in BERTScore suggests stronger semantic agreement with expert reference responses. These findings indicate that combining local retrieval, biomedical literature retrieval, and supplementary contextual evidence provides more effective grounding than relying on a single retrieval source.

VI. RESULTS

A. Ablation Study Results

Table II summarizes performance across the three retrieval configurations.

The results indicate that retrieval contributes positively to biomedical response quality relative to generation without external evidence. The Baseline configuration produced lower lexical and semantic agreement with reference responses, suggesting that retrieval grounding improves response consistency.

The RAG-Only configuration achieved the highest BLEU-1 score, indicating stronger lexical overlap with reference answers [16]. This suggests that local semantic retrieval improves surface-level alignment between generated and expected responses.

The Full Multi-Agent configuration achieved the highest ROUGE-1 and BERTScore F1 scores [17], [19], indicating improved content coverage and stronger semantic agreement with expert references. These findings suggest that combining

TABLE II
ABLATION STUDY RESULTS ON THE PUBMEDQA EVALUATION SET

Configuration	BLEU-1	ROUGE-1	BERTScore F1
Baseline (No Retrieval)	0.2769	0.2902	0.8748
RAG Only	0.3455	0.3530	0.8713
Full (Multi-Agent)	0.2895	0.5789	0.9201

TABLE III
FULL MULTI-AGENT CONFIGURATION — COMPLETE METRIC SUITE

Metric	Score
BLEU-1	0.2895
ROUGE-1	0.5789
METEOR	0.7117
BERTScore F1	0.9201

local retrieval, biomedical literature retrieval, and supplementary contextual evidence provides more effective grounding than relying on a single retrieval source.

Although evaluation was conducted at prototype scale on the PubMedQA benchmark [10], the observed trend supports the hypothesis that heterogeneous retrieval improves biomedical response quality.

B. Full Configuration Metrics

Table III reports the complete evaluation metrics for the Full Multi-Agent configuration.

The final configuration maintained strong semantic alignment while preserving competitive lexical performance. In particular, the high METEOR and BERTScore values [18], [19] indicate that generated responses remained semantically faithful even when wording differed from the reference answers.

This behavior is particularly desirable in biomedical question answering, where preserving factual meaning is often more important than reproducing exact wording.

C. System Demonstration

To qualitatively demonstrate system behavior, Figures 2 and 3 present representative outputs from MediQuery.

Figure 2 illustrates the interactive interface used for biomedical question answering and evidence exploration. The interface integrates natural language querying, multi-source retrieval, grounded response generation, and citation-aware evidence presentation within a unified workflow.

Figure 3 presents the dynamically generated biomedical knowledge graph constructed from retrieved evidence. The graph enables visualization of extracted entities and relationships, providing an interpretable view of contextual information supporting generated responses.

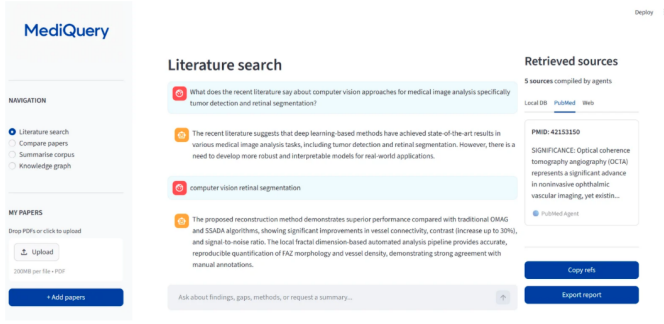


Fig. 2. MediQuery user interface demonstrating grounded biomedical response generation and evidence presentation.

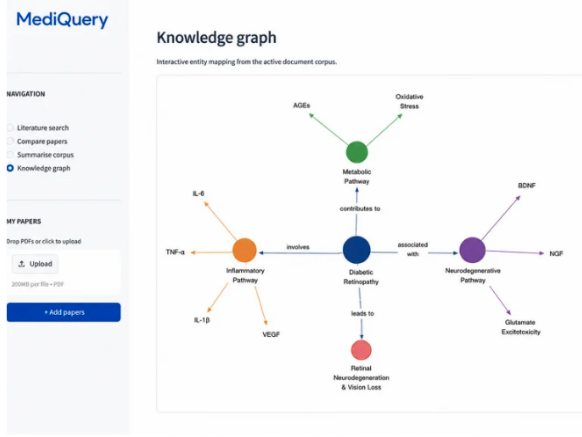


Fig. 3. Dynamic biomedical knowledge graph generated from retrieved evidence and used for contextual augmentation.

VII. DISCUSSION

A. Multi-Agent Retrieval Improves Semantic Grounding

The Full Multi-Agent configuration achieved the strongest semantic performance across evaluation metrics, particularly in ROUGE-1 and BERTScore F1 [17], [19]. These results suggest that integrating heterogeneous evidence sources—including local retrieval, biomedical literature retrieval, and supplementary contextual information—improves alignment with expert reference responses.

The strong METEOR performance [18] further indicates that generated responses preserve semantic meaning beyond exact lexical overlap, capturing related terminology and variations in biomedical language.

B. BLEU-1 Trade-off: Precision versus Context

A notable observation is that the Full Multi-Agent configuration did not achieve the highest BLEU-1 score despite producing the strongest semantic performance. We attribute this behavior to the broader evidence context available during generation.

Additional retrieval sources encourage the model to produce richer and more explanatory responses, which may reduce exact lexical overlap with the relatively concise benchmark references. This observation highlights a known limitation of

lexical metrics for evaluating long-form biomedical generation [16] and reinforces the importance of complementary semantic evaluation measures [19].

C. RAG-Only Retrieval Shows Mixed Performance

The RAG-Only configuration achieved stronger lexical overlap but lower semantic alignment compared with the Full Multi-Agent configuration. This suggests that retrieval from a static local index alone may improve wording similarity without necessarily improving contextual completeness.

The inclusion of external biomedical literature and supplementary contextual retrieval appears to provide additional semantic depth that improves overall response quality.

D. Contribution of Domain Adaptation

Fine-tuning contributed to adapting the generative model toward biomedical terminology and response structure while maintaining efficient deployment characteristics. Although training was intentionally lightweight for prototype development, the results indicate that parameter-efficient adaptation can support domain specialization without requiring full-scale retraining [9].

E. Limitations

Several limitations should be considered when interpreting the reported findings. First, evaluation was conducted at prototype scale using a limited subset of benchmark samples, and larger-scale evaluation is necessary before drawing broader conclusions. Second, the current retrieval pipeline executes sequentially, which increases response latency and may limit scalability under larger workloads. Third, fine-tuning and retrieval remain dependent on available biomedical data and may not generalize equally across emerging or underrepresented medical topics. Finally, although retrieval grounding reduces unsupported outputs, automated hallucination verification and stronger evidence-validation mechanisms remain areas for future improvement.

VIII. FUTURE WORK

Multiple opportunities exist to extend MediQuery beyond the current prototype implementation. Future work will focus on improving system efficiency through parallel multi-agent execution to reduce retrieval latency and enable more responsive evidence aggregation. Additional retrieval guardrails and confidence-aware generation strategies will be explored to further reduce unsupported outputs and improve robustness for queries involving limited or emerging biomedical evidence. From a modeling perspective, future work will investigate extended domain adaptation and larger-scale fine-tuning to improve biomedical reasoning and response calibration. Evaluation will be expanded to larger benchmark subsets and supplemented with retrieval-oriented and human-centered assessment methods to provide a more comprehensive understanding of system performance and reliability. Future iterations will also explore scalable deployment architectures and stronger biomedical entity extraction methods to improve knowledge graph quality and support richer multi-hop evidence reasoning.

IX. CONCLUSION

This work presented MediQuery, a domain-specific multi-agent Retrieval-Augmented Generation system for biomedical research assistance. By combining local retrieval, biomedical literature retrieval, and evidence-aware generation, MediQuery improves response grounding, citation transparency, and access to current biomedical knowledge.

Experimental results indicate that integrating heterogeneous retrieval sources improves semantic alignment compared to generation without retrieval support. Overall, MediQuery demonstrates the potential of multi-agent retrieval and grounded generation for more reliable and interpretable biomedical question answering.

REFERENCES

- [1] C. Tenopir, R. Volentine, and D. W. King, "Article and journal value in the digital age," *Learned Publishing*, vol. 28, no. 1, pp. 31–41, 2015.
- [2] athenahealth, "Physician sentiment survey," Research Report, 2024.
- [3] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.
- [4] S. Barnett, C. Toro, D. Dowe, and C. Kemp, "Seven failure points when engineering a retrieval augmented generation system," arXiv:2401.05856, 2024.
- [5] RTInsights, "Real-time RAG pipelines: Achieving sub-second latency in enterprise AI," Research Report, 2026.
- [6] Anthropic, "Contextual retrieval," Technical Report, 2024.
- [7] Y. Gu et al., "Domain-specific language model pre-training for biomedical natural language processing," *ACM Transactions on Computing for Healthcare*, vol. 3, no. 1, pp. 1–23, 2021.
- [8] M. Neumann, D. King, I. Beltagy, and W. Ammar, "ScispaCy: Fast and robust models for biomedical natural language processing," in *Proc. BioNLP Workshop*, pp. 319–327, 2019.
- [9] E. J. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.
- [10] Q. Jin et al., "PubMedQA: A dataset for biomedical research question answering," in *Proc. EMNLP-IJCNLP*, pp. 2567–2577, 2019.
- [11] D. Gupta, D. Bartels, and D. Demner-Fushman, "A dataset of medical questions paired with automatically generated answers and evidence-supported references," *Scientific Data*, vol. 12, no. 1, p. 1035, 2025.
- [12] LangChain, "LangGraph: Stateful, multi-actor applications with LLMs," Documentation, 2024.
- [13] Qdrant, "Qdrant: Vector database for AI applications," Documentation, 2024.
- [14] D. Han and M. Han, "Unsloth: Faster and memory-efficient LLM fine-tuning," GitHub Repository, 2024.
- [15] mtRAG Research Group, "Multi-turn retrieval-augmented generation for long-context medical QA," MIT Press, 2025.
- [16] K. Papineni, S. Roukos, T. Ward, and W. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. ACL*, pp. 311–318, 2002.
- [17] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Proc. ACL Workshop*, pp. 74–81, 2004.
- [18] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proc. ACL Workshop*, pp. 65–72, 2005.
- [19] T. Zhang et al., "BERTScore: Evaluating text generation with BERT," in *Proc. ICLR*, 2020.

APPENDIX A TECHNICAL APPENDIX

This appendix provides supplementary mathematical definitions, implementation details, software descriptions, deployment information, and technical terminology used throughout MediQuery.

A. Evaluation Metric Definitions

BLEU-1

BLEU evaluates lexical precision using unigram overlap.

$$BLEU = \frac{\text{Matched unigrams}}{\text{Generated unigrams}}$$

Higher BLEU indicates stronger lexical agreement.

ROUGE-1

ROUGE evaluates recall-oriented content overlap.

$$ROUGE = \frac{\text{Matched unigrams}}{\text{Reference unigrams}}$$

Higher ROUGE indicates broader content coverage.

Recall

$$Recall = \frac{TP}{TP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

F1 Score

$$F_1 = 2 \cdot \frac{Precision \times Recall}{Precision + Recall}$$

METEOR

$$METEOR = \frac{10PR}{R + 9P}$$

where:

$$P = Precision, \quad R = Recall$$

BERTScore

$$BERTScore = \frac{1}{N} \sum_i \max_j \cos(h_i, \hat{h}_j)$$

where:

- h_i = generated contextual embedding
- $\hat{h}_j$ = reference contextual embedding

Higher BERTScore indicates stronger semantic agreement.

B. Retrieval and Embedding Definitions

Cosine Similarity

$$Similarity = \frac{A \cdot B}{|A||B|}$$

Used to compare semantic vector similarity.

Mean Pooling

$$E = \frac{\sum h_i m_i}{\sum m_i}$$

where:

- h_i = token embedding
- m_i = attention mask

Top-k Retrieval

$$R = TopK(Similarity(q, d))$$

where:

- q = query vector
- d = document vector

Chunk Generation

$$Chunks = \frac{DocumentLength}{ChunkSize}$$

Chunking improves indexing scalability and retrieval efficiency.

C. Software Stack and Implementation Details

Streamlit

Streamlit was used as the frontend application framework for MediQuery. It enabled construction of an interactive biomedical question-answering interface supporting natural language querying, PDF uploads, evidence exploration, and knowledge graph visualization.

Docker

Docker was used to containerize ingestion, indexing, retrieval, and deployment services. Containerization simplified dependency management and ensured reproducible execution environments across local and deployment workflows.

Apache Kafka

Apache Kafka was integrated as the asynchronous ingestion and event-streaming layer. Kafka decoupled ingestion from indexing and embedding generation, improving throughput and allowing ingestion workloads to remain durable under temporary downstream delays.

QdrantDB

QdrantDB served as the persistent vector database for semantic retrieval. Biomedical document chunks were embedded and indexed into vector collections to support low-latency similarity search during inference.

LangGraph

LangGraph coordinated the multi-agent execution workflow. Retrieval, synthesis, and generation stages were represented as structured execution graphs with controlled state transitions.

PubMedBERT

PubMedBERT was used as the embedding backbone for retrieval. Since it was pretrained exclusively on biomedical literature, it provided stronger domain alignment than general-purpose encoders.

LLaMA-3

LLaMA-3 served as the generative backbone for biomedical response synthesis and local experimentation. Parameter-efficient adaptation allowed specialization without full retraining.

AWS Bedrock

AWS Bedrock was integrated for production-grade model serving and inference. Bedrock simplified access to foundation models without requiring dedicated infrastructure management.

Amazon Nova Pro Lite

Amazon Nova Pro Lite was selected as the deployment inference model due to lower latency and faster response generation. The model provided improved responsiveness for real-time biomedical interaction.

SciSpaCy

SciSpaCy supported biomedical entity extraction and contributed to dynamic knowledge graph construction.

NetworkX

NetworkX managed graph construction and relationship representation across biomedical entities.

Hugging Face Evaluate

The Evaluate framework was used to compute BLEU, ROUGE, METEOR, and BERTScore consistently across experiments.

D. Deployment Architecture

The MediQuery deployment pipeline combines local retrieval and cloud-based generation.

- Docker containers isolate ingestion and retrieval services.
- Kafka provides asynchronous ingestion and throughput decoupling.
- QdrantDB stores semantic embeddings.
- AWS Bedrock serves production inference.
- Nova Pro Lite reduces latency.
- Uploaded PDFs are processed independently from the biomedical corpus.

Pipeline summary:

[500 papers $\rightarrow$ 8500 + indexed chunks]

E. Computational Characteristics

Approximate complexity estimates:

[Embedding: $O(n)$]

[VectorSearch: $O(\log N)$]

[Generation: $O(T^2)$]

where:

- n = token count
- N = index size
- T = sequence length

F. Technical Glossary (Alphabetical)

Chunking — Division of long documents into smaller retrieval units.

Dense Embedding — Continuous vector representation of text.

Fault-Tolerant Deserialization — Recovery from malformed records and noisy PDFs.

Grounding — Constraining generation using retrieved evidence.

Latency — End-to-end response time.

Multi-Agent Retrieval — Evidence retrieval using multiple specialized agents.

Prompt Synthesis — Aggregation of retrieved evidence before generation.

Semantic Retrieval — Retrieval based on contextual similarity.

Throughput Decoupling — Separation of ingestion and indexing workloads.

Vector Index — Data structure optimized for similarity search.

Zero-Shot Generation — Generation without retrieval or adaptation.